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FISH 460
29 May 2026

Into the Crabrooms

Abstract

The invasive European green crab (*Carcinus Maenus*) is a problem for our coastal estuaries. They outcompete and predate on native shore crabs (*Hemigrapsus*). Trapping has become the main mode of removal for European green crabs but how does this affect the stress of native shore crabs? Two different trap types were used for this experiment to determine if there was a better option for the native shore crabs. Researchers constructed makeshift Fukui wire traps and crab slabs, an innovative trap being tested in the Tulalip tribe. Native shore crabs were placed in traps with and without the presence of a predator. They were surveyed over three weeks and tested for righting time, respiration and protein levels. Overall, there was no difference in righting times, protein assay, or respiration between the two trap types, with or without the presence of a predator. Testing in real traps would need to be done in order to confirm if there are physiological consequences to native shore crabs. Protecting our native species is extremely important as the rise of invasive species and changing ocean conditions threaten their survival. Without these species, local ecosystems would be disrupted and the consequences would spread well beyond their oceanic environment.

Introduction

Shore Crabs (*Hemigrapsus*) are small crabs native to the North American Pacific Coast. These crabs are very small and are great osmoregulators (Cowles, 2005). The most common Shore Crabs in the Puget Sound are the Hairy shore crab (*Hemigrapsus Oregeonensis*) and the Purple shore crab (*Hemigrapsus Nudus*) (Cowles, 2005). These crabs are similar physiologically and have similar ecological niches. There are also other species that occupy this niche as well. The European Green Crab (*Carcinus Maenus*) is an invasive crab species that is native to European coast lines and infiltrated Atlantic waters in the 1800s (Klassen et al, 2004). They are known as one of the top 100 most invasive species in the world because of their innate ability to survive even the most harsh conditions (Ens et al., 2022). European green crabs destroy eelgrass and benthic environments by eroding the substrate. They are also very destructive to bivalve fisheries and native crab populations because they are both predators and competitors (Ens et al., 2022). European green crabs have begun to invade the Puget Sound, however it is only in the beginning stages (Cauvel, 2024). Efforts to remove European green crabs once they have settled in an environment have all failed due to their high levels of reproduction and planktonic larval phase (Klassen et al, 2004). However, there are still efforts to contain the populations which can aid in the protection of local-scale marine environments (Flynn et al, 2024). The most common form of removal efforts are traps, specifically the Fukui trap (Ens et al., 2022). Fukui traps are wire traps that are baited and allow crabs to climb in naturally. These traps are used for many types of crabs and are not specifically made for trapping European green crabs. The other type of

trap that is being tested is called a crab slab (Cauvel, 2024). This “trap” is more of a constructed ideal environment for the European green crab. Instead of trapping the crab, the slab is open and allows them to move in and out freely. The hope is that European green crabs will settle in the slab and allow for easy removal when a person comes to collect it (Cauvel, 2024). This removal strategy is currently being tested in the Tulalip tribe and is one of the only removal methods focused specifically on European green crabs in the Puget Sound (Cauvel, 2024). The only problem with both of these traps is that they have no protections for native crab populations. There have been studies that showed that crabs in traps experience negative behavioral and reproductive changes (Darnell et al., 2010, Archdale, 2007) but they do not use native shore crabs as a key specimen. This gap in the literature was the basis for this experiment. This experiment tested if there was a difference in stress of native shore crabs between the traditional Fukui traps and the newly developed crab slab when aiding in European green crab capture. Since crab slabs mimic the natural environment, it was thought that this trap would cause less stress on native species than the wire Fukui traps. The presence of predators is also very common in a trap in the wild. This presence would also cause an increase in stress, however it would decrease in the crab slab because of the mimicked environment. By diving into this phenomenon, researchers hope to be able to protect native populations while also fighting the surge of European green crabs in the Puget Sound.

Methods

The experiment contained four traps in total; two Fukui (wire) traps and two crab slabs. One set of each trap would be in the presence of a European green crab in order to simulate the stress of being trapped with a predator. Both *Hemigrapsus Oregeonensis* and *Hemigrapsus Nudus* were used in this experiment. Shore crabs and European green crabs were collected by Andy Nutzhorn, the lab manager. Researchers used 12 shore crabs and two European green crabs for this experiment. The crabs were all male and roughly the same size which was determined by weight. Physiological and behavioral tests were only run on shore crabs. The first week's data was collected before the crabs were exposed to any treatment. The experiment lasted a total of three weeks in which crabs were starved for the full duration.

Traps and Holding tanks

Since our space was limited in the lab, makeshift traps were created in small cylindrical jars. Two wire traps were created using a flexible, mesh net. The inside of the jar was lined with a net connected together by fishing line. The crab slabs were made using black out tape and the inside of the jar had a layer of sediment on the bottom. Three shore crabs were placed in each makeshift trap. One larger holding container held one wire trap and one crab slab, while the other larger holding container held the additional two traps along with a European green crab. Both holding containers were filled with salt water and placed in a larger water bath to control temperature. Containers were cleaned and water was changed weekly.

Righting time

Before each weekly test, crabs were placed on their carapaces and timed to see how long it took to right themselves. Righting time was used as a proxy for health (Wilson et al., 2022).

Resazurin

Resazurin was used as a way to measure respiration. The resazurin mixture was created in accordance with a resazurin calculator (Huffmyer et al.). Every week, each crab was placed in a well plate filled with 13 ml of resazurin. The plate was covered and they were left for one hour. 200 µl of resazurin were taken at times 0, 30, and 60 and placed into a 96 well plate. After the full hour, crabs were washed and returned back to their containers. The well plate was then analyzed for fluorescence to determine the total respiration. Resazurin was used to test stress as a crab that is experiencing more stress will have a change in their total respiration (Bao, 2019).

Hemolymph Protein Assay

In the final week of the experiment, hemolymph was extracted from each shore crab to assess their protein levels. Hemolymph was extracted through the joint between the legs and either the 4th or 5th abdominal segment. A needle was placed into the segment and a small amount of hemolymph was taken. Hemolymph was placed into microcentrifuge tubes and then deposited into a 96 well plate. The samples were then separated into two distinct samples, diluted and saturated. These samples were tested for absorbance which gave us a proxy for protein levels. When stressed, Crustaceans protein expression and metabolism will change (Hapsari et al., 2025) so by analyzing the protein levels, it would be possible to determine if the treatments caused stress on the crabs.

Results

Righting Time

The crabs in the crab slab treatment showed an increase in righting time over the three weeks. There is almost no change between weeks one and two however in week three there is a large increase in righting time from 1 second to around 4 seconds. The crab slab with predator present showed an opposite trend. In week one they had the highest righting time around 3 seconds then decreased to less than 1 second by week three. The wire trap treatment also has a decrease in righting time over the three weeks however it is not as drastic as the crab slab with predator. Righting time started at about 1.5 seconds and decreased to less than 1 second by week three. The wire trap with predator showed an increase over the weeks as well, however the median time remains similar. The median began below 1 second in week one and increased slightly to around 1.5 seconds in week three. During the testing in week three, there was an individual who skewed the data drastically which accounts for the large box. The p-value from a Kruskal-Willis test was 0.4704.

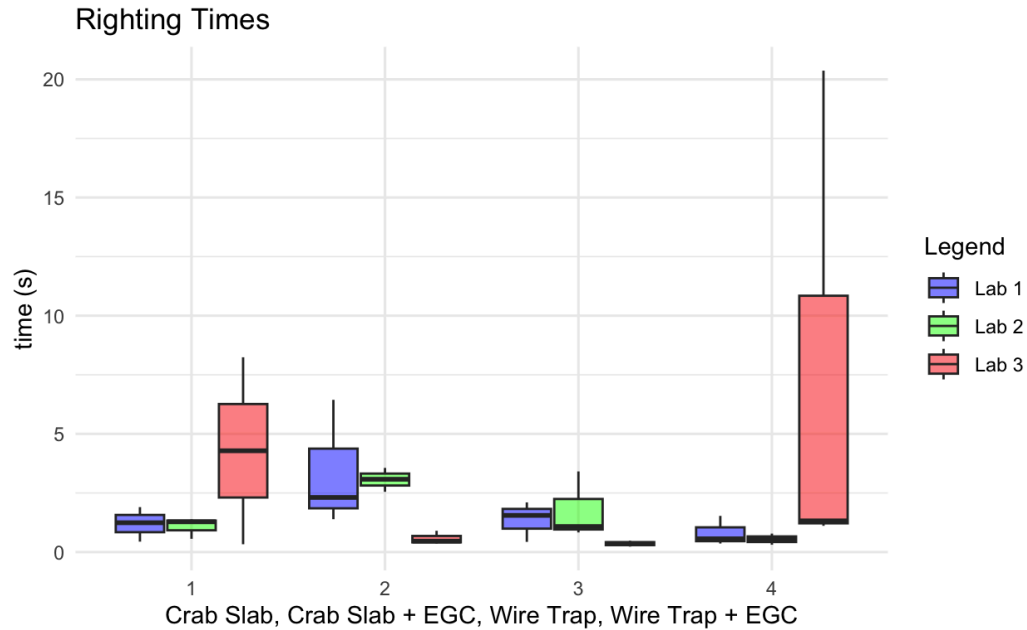


Figure 1 Righting time (in seconds) of crabs across treatments. Each color represents a different week they were tested. Groups of weeks are separated into treatments.

Respiration

For the crabs in the crab slab with no predator treatment, respiration sits around 500 fluorescence. The crab slab with predator has a very similar median, minimum and maximum value to the crab slab without predator. Wire trap, no predator treatment has the greatest level of respiration with a median around 1000 fluorescence. The wire trap with predator treatment has the least amount of respiration sitting at less than 500 fluorescence. The wire traps have the greatest respiration difference. The p-value from a Kruskal-Willis test was 0.781.

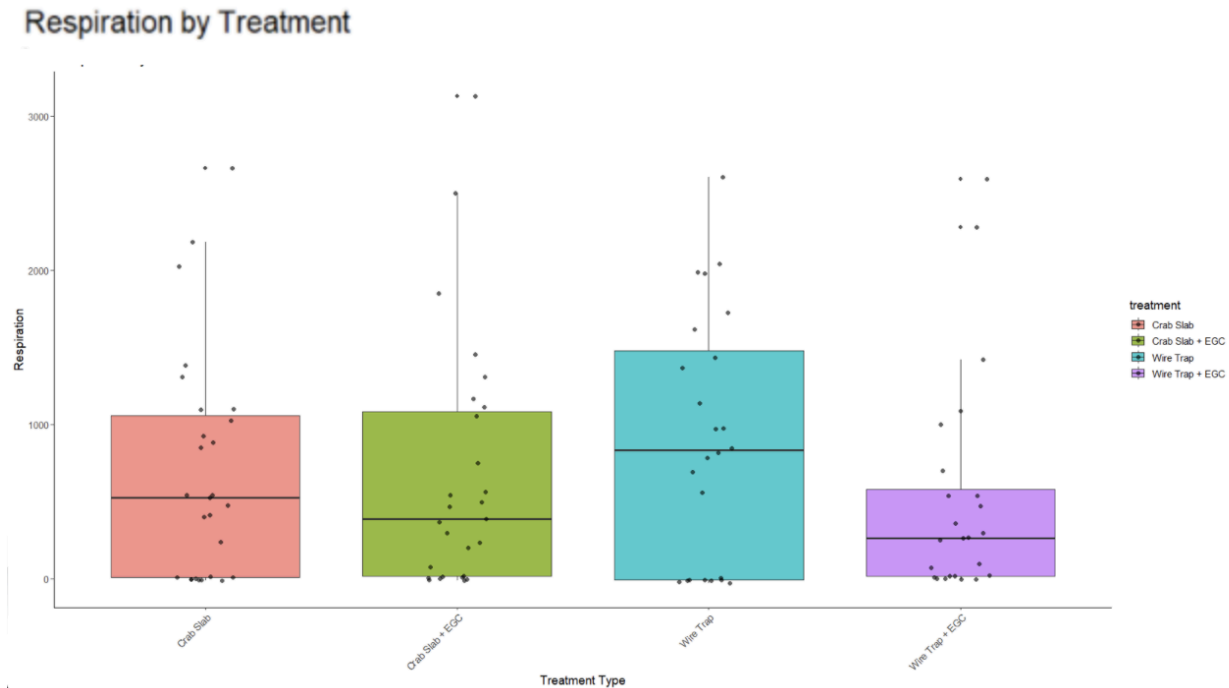


Figure 2. The figure shows the respiration level of each crab group by treatment type over the three weeks of the experiment. The colors represent each treatment and the points represent a single crabs respiration at one time point during the respiration test.

Protein Assay

The crab slab, no predator treatment has two individuals that were able to be tested for protein assay. The first individual shows a protein level absorbance of 2 while the second individual shows an absorbance of around 3.25. The crab slab with predator treatment also shows two individuals, however one only has a value for the diluted assay. The first individual shows a protein level around 3 while the second individual shows a diluted protein level just above 3.75. The wire trap, no predator treatment shows two individuals as well with similar numbers to the individuals in the crab slab, no predator. The first individual has a protein concentration around two while the other has a protein level just below 3.75. The final treatment, wire trap with predator only has one individual. This individual has a protein concentration around 3.0. The p-value from a Kruskal-Willis test was 0.838.

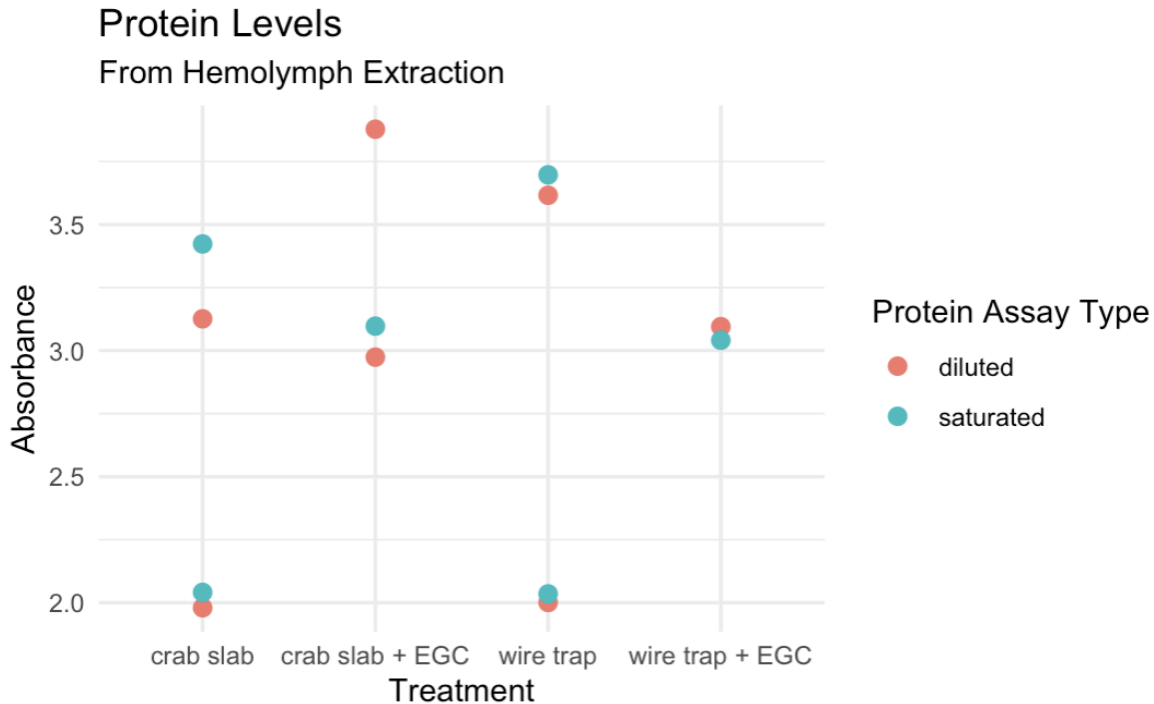


Figure 3 This figure shows protein absorbance from hemolymph extractions over the different treatments. Colors represent different assay types. Each combination of two assay types shows individual crabs. Overflows were omitted from the data and considered to be N/A.

Discussion

The results of this experiment were different than predicted. Health (righting time) was only negatively affected by the crab slab with no predator. It was expected that righting time would decrease in wire traps and in the presence of predators due to a decrease in health, however this was not the case (Wilson et al., 2022). Instead, righting time seemed random and did not follow a specific pattern for any of the treatment besides the crab slab. Even though there are some patterns within the data, the p-value showed that health was not correlated with treatment. The protein assay also did not show any significant patterns. There was a similarity between protein levels in individuals not exposed to predators, however there is not enough evidence provided in our predator datasets to prove a correlation is occurring. It is possible the protein assay was skewed because crabs were starved and therefore not able to produce large amounts of protein (Mente et al., 2010). The respiration also showed no significant difference between treatments based on the p-value. For respiration, there was a slight difference between in the wire trap with and without the presence of a predator but this is not mirrored in the crab slab. Crabs are very successful osmoregulators which means that when they are stressed they can switch to anaerobic respiration in order to limit their physical exertion (Bao, 2019). This could be what is happening in the crab slab with predator treatment as it has the lowest respiration rate. Based on these results, there is not an impact on health, protein levels or respiration in different traps in the presence or absence of predators. In the future, more testing in real traps will be

needed to determine if there are chronic long term issues that have a negative impact on the physiology of native shore crabs.

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